

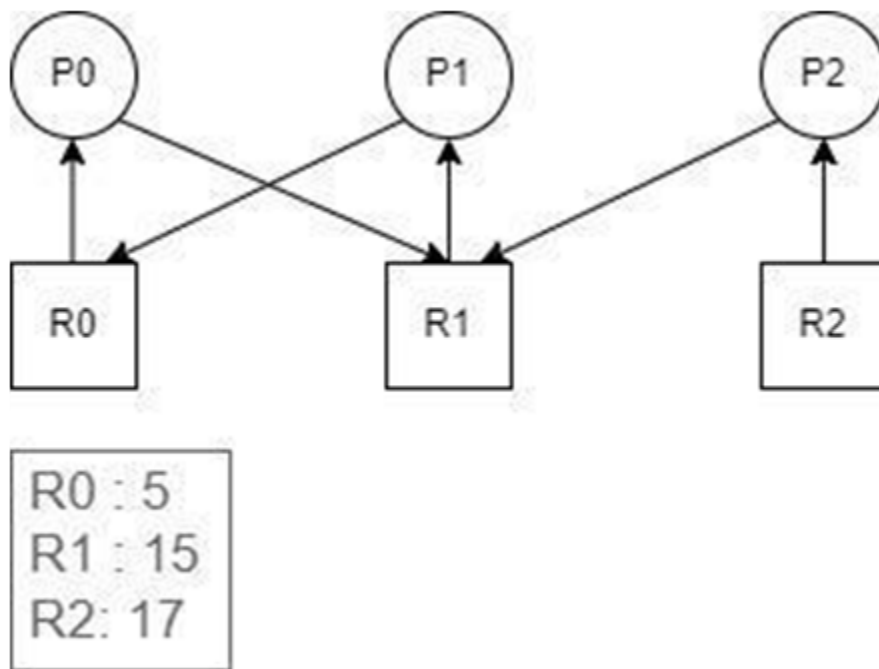
Short Question:

1. Describe the different types of Operating System structures with examples.
2. What is the difference between kernel mode and user mode?
3. What do you mean by a critical section in process synchronization?
4. A shared printer queue is accessed by multiple processes in a time-sharing system. What would happen if:
a) Two processes simultaneously modify the queue without mutual exclusion?
b) The order of execution changes every time the program runs?
5. What is Round Robin CPU scheduling? Mention one advantage and one disadvantage.
6. Define a process in an operating system. Explain the process life cycle with a state diagram. Also, briefly describe the role of the Process Control Block (PCB) in process management.
7. In a system, the processes are $[p_0, p_1, p_2]$, the max vector of the single resource is $[10, 4, 9]$ and Allocation vector is $[5, 2, 2]$, whereas the total number of instances is 13. Is $\langle p_0, p_2, p_1 \rangle$ a safe sequence? Justify!
8. Define system calls. Mention any two types of system calls.
9. If an OS supports **4 types of system calls** and each type has **5 system calls**, calculate the total number of system calls.
10. A computer system takes 4 seconds for BIOS execution and 6 seconds for OS loading. Calculate the total boot time.
11. What is a thread? Mention any two benefits of multithreading.

12. Define race condition.
13. Define semaphore.
14. Consider there are two processes P0 and P1 having two statements s0 and s1 respectively. Implement a semaphore to synchronize the processes in such a way that P0 will be executed on completion of P1.
15. What is a deadlock? Explain the conditions necessary for a deadlock to occur.
16. Explain deadlock detection and recovery in operating systems.
17. What is deadlock avoidance? Mention the role of Banker's algorithm.
18. Explain deadlock prevention with any one example.
19. Explain the concept of files and file attributes.
20. Describe directory structures and their implementation.
21. If the main memory has **1024 frames** and each frame size is **4 KB**, calculate the total main memory size.
22. A process generates **100 page references** and causes **20 page faults**. Calculate the page fault rate.
23. A system has a page size of 4 KB. How many pages are required for a process of size 20 KB?
24. Explain Inter-Process Communication (IPC) and its importance in operating systems.
25. What is paging?
26. What is segmentation?

27. Define virtual memory. Briefly explain the term “Copy-on-Write” (COW) for Virtual Memory Management.
28. What is demand paging?
29. Describe thrashing and its impact on system performance.
30. What do you mean by swapping in memory management?
31. What is RAID? State its advantages in storage systems.

Long Question:



1. Consider the RAG and resource enumeration number as given in the box and each resource having a single instance. Consider that system implements a deadlock prevention technique. Verify that the above diagram is valid or not. Modify the RAG and draw the possible correct RAG.(if required)
2. Describe the concept of invalid pages and discuss the same with an example.

3. The following information is stored in the segment table in the following format, <segment number, base, limit>. The content of the segment table in each row is as follows - <0, 219, 600>, <1, 2300, 14>, <4, 1952, 96>. Calculate the physical address for the logical address (0,300).
4. Define contiguous memory allocation. Explain its working in operating systems and mention its advantages and disadvantages.
5. Consider a system where, Process={p0,p1,p2,p3}, Arrival time={0,1,2,0} burst time={4,2,3,2}, In SJF method CPU scheduling of uni-processor system, which process is executed after p3?
6. A system has 100 KB of main memory. Three processes require memory blocks of 20 KB, 35 KB, and 15 KB. Show how these processes would be allocated using contiguous memory allocation. Indicate any external fragmentation after allocation.
7. Consider a disk scheduler that implements SCAN, C-LOOK having a sequence of cylinders as follows: 51,56,44,89,34,23,5,98,77. A disk system has 100 cylinders (0–99). Calculate the total amount of cylinder head seek. Consider the starting head at 54.
8. Distinguish between preemptive and non preemptive CPU Scheduling algorithms. How does the Round-Robin CPU scheduling algorithm work? Consider the following set of processes, with the arrival times and the CPU-burst times given in milliseconds.

Process	Arrival Time	Processing Time
P1	0	5

P2	1	3
P3	2	4
P4	4	6

Calculate average waiting time and average turnaround time for these processes using the Round-Robin algorithm with a time quantum of 2 ms.

9. Consider the following set of processes with their Arrival Time (AT) and Burst Time (BT):

a) Schedule the processes using Shortest Job First (Non-Preemptive) algorithm.

b) Draw the Gantt chart.

c) Calculate the average waiting time and average turnaround time.

10. In a system, the following parameters are known: Disk transfer rate = 120 MB/s, Disk latency = 10 ms. Two processes of sizes 40 MB and 90 MB are candidates for swapping. a) Identify which process should be swapped out first and justify your answer.

→ Calculate the swapping time for the chosen process.

→ Discuss one limitation of swapping highlighted by this example.

11. For the RR scheduling algorithm, Burst time of five processes (p0,p1,p2,p3,p4) are as follows (2ms, 3ms, 15ms, 1ms, 2ms).

Estimate the most suitable time quantum which would optimize

the throughput as well as average turnaround and average wait time.

12. A system has 5 disk requests pending in the request queue:

98, 183, 37, 122, 14

The initial position of the disk head is at 53.

- a) Calculate the total head movement using the FCFS disk scheduling algorithm.
- b) Calculate the total head movement using the SSTF disk scheduling algorithm.
- c) Compare the results and identify which algorithm gives better performance.

13. consider the following code snippet and answer the following questions:

```
main()
{
    printf("\n Hi..");
    fork();
    printf(" Hello..");
    fork();
    printf(" My dear students...");
}
```

Calculate the number of processes will be created and discuss the parent-child relationship among them.

14. Explain the concept of page replacement in operating systems. Describe FIFO, LRU, and Optimal Page Replacement algorithms, including their advantages and disadvantages.

15. Consider the following sequence of memory references generated by a single program in a pure paging system: 10, 11, 104, 170, 173, 177, 309, 245, 246, 247, 458, 364. Determine the no. of page faults for each of the following page replacement policies. Assuming three (3) page frames are available and all are initially empty. The size of a page is 100 words: (i) FIFO (ii) LRU and (iii) Optimal page replacement algorithm.

16. Consider the following allocations, request and claim of the multiple resources(all are single instance) by multiple processes in the system are represented as matrix as follows:

Allocation	R0	R1	R2
P0	1	0	0
P1	0	0	0
P2	0	1	0
P3	0	0	1

Request	R0	R1	R2
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P0	0	1	1
P1	1	1	0
P2	0	0	0
P3	1	0	0

Claim	R0	R1	R2
P0	0	0	0
P1	0	0	0
P2	1	0	0
P3	0	1	0

Draw the RAG and check the Unsafe / Safe state. Identify the particular claim/s which might cause the system to be unsafe. Give your justification that a safe state never leads to deadlock.

17. Explain the Banker's Algorithm for deadlock avoidance, including the purpose, data structures (Allocation, Maximum, Need, Available), and the concept of safe state.

A system has **5 processes (P0–P4)** and **3 resource types (A, B, C)** with the following details.

Total instances of resources:

A = 10, B = 5, C = 7

Allocation Matrix

Process	A	B	C
P0	0	1	0
P1	2	0	0
P2	3	0	2
P3	2	1	1
P4	0	0	2

Maximum Matrix

Process	A	B	C
P0	7	5	3
P1	3	2	2
P2	9	0	2
P3	2	2	2
P4	4	3	3

- Calculate the **Available** resource vector.
- Compute the **Need** matrix.
- Determine whether the system is in a **safe state**.
- If the system is safe, find a **safe sequence** using Banker's Algorithm.